

Assessment #2
Annotated Bibliography Assessment

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Topic: PCB Design

Key: (1) Citation, (2) Introduction, (3) Aims & Research Methods, (4) Scope, (5) Usefulness, (6) Limitations, (7) Conclusion, (8) Reflection

(1) Yang, Paige. "PCB Thermal Management | PCBDesign Lab." *PCBDesign Lab*, 22 July 2026,

<https://www.pcbdesignlab.com/pcb-thermal-management/>. Accessed 11 Sept. 2026.

(2) This article explains how PCB designers manage excess heat with copper, heatsinks, airflow, and component placement, and explains concepts such as thermal resistance and junction temperature and how these affect the PCB's reliability. (3) The article's purpose is to explain how to manage heat as the PCB is being designed instead of trying to solve it when it becomes a problem, which it does by using engineering equations, technical standards, and examples to explain different thermal control methods.. (4) The article mainly focuses on concepts like temperature, power dissipation, thermal resistance, copper area, thermal vias which are important since they actually tell us how effectively heat is being pulled from the PCB into the environment.. (5) This source is helpful because it showed me that even with correct theoretical electrical design, poor thermal design can cause unwanted problems and also made me realize that the PCB itself could be used to remove heat from its components further showing how important PCB design is..(6) Although the article is very detailed and uses technical language, its primary purpose is to act as an engineering guide rather than real research, so its recommendations might not be universal, and some of the examples might actually produce different results due to factors like material, components, airflow, and operating conditions. (7) From this source, I can use this knowledge to look deeper into how different PCB designs can improve heat dissipation and possibly test how changes in copper area or thermal vias affect temperature, connecting to a previous interest of mine in heat dissipation produced by microcontrollers. (8) This source is very relevant to my learning because it shows that thermal management is an extremely important part of designing PCBs that are reliable and connects to my previous experiences relating to microcontrollers and thermal design within them.

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(1) Peterson, Zachariah. "PCB Routing Guidelines for Altium Designer." *Altium*, 17 Sept. 2021, <https://resources.altium.com/p/pcb-routing>. Accessed 11 Sept. 2026.

(2) This article explains how traces should be routed on PCBs and how decisions regarding this routing can affect the board's actual performance and signal integrity, and it covers both basic routing concepts and more advanced ones commonly seen in high-speed circuits. (3) The purpose of this article is to teach designers like me how to actually route traces properly by explaining concepts such as trace width, impedance, spacing, vias, and signal paths, and all of this information is shown through engineering principles, examples, and common PCB design practices. (4) The article covers basic topics like trace width, impedance, and spacing all the way to multilayer routing and differential pairs; these variables are discussed thoroughly as they can affect signal quality, interference, current capacity, and the overall reliability of a circuit. (5) This article is beneficial as it helped me understand that routing components together is much more complicated than just drawing a line between two different components to connect them with a trace. I actually found it super interesting that the exact shape, length, and angles on a trace can actually affect how signals behave, and I want to look more into other factors of tracing that can influence this. (6) A limitation of this article is that they cover lots of topics, but they do so broadly without going into the experimental effects of each routing method; since it's published by Altium, some information is also shown with their own PCB design software rather than as independent research. (7) In conclusion, I will apply this information by studying and learning how different routing choices can actually affect circuit performance and using my findings when designing my own PCB, giving me a potential Research direction I want to explore. (8) This is directly relevant to my ISM topic since routing is an important skill I have to learn and understand to go from simply understanding circuits to being able to actually design an effective, reliable PCB.

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(1) Keim, Robert. "Guide to PCB Design: From PCB Schematic to Board Layout."

Allaboutcircuit, All About Circuits, 28 Aug. 2018,

<https://www.allaboutcircuits.com/technical-articles/guide-to-pcb-design-pcb-schematic-board-layout/>. Accessed 11 Sept. 2026.

(2) This article explains the process of turning a circuit design on a computer into an actual physical PCB and eventually preparing it for manufacturing, providing an overview of the actual steps of turning an idea into an actual board. (3) The article aims to provide a practical guide for designers to materialize their PCBs. The author does this by using personal design experience, examples, and explanations of PCB software and processes. (4) Under the scope of this article are schematics, footprints, component placement, traces, vias, design-rule checks, and manufacturing preparation. These areas are important because any mistakes in them can cause a PCB that might work fine in software to fail in the real world. (5) This article helped me understand the complete design process and not just specific parts of it, giving me a better idea of what skills I need to learn and develop before I can successfully design a PCB myself. (6) Some limitations are that the article was published in 2018, which means some of its software information and design practices might be outdated in the current state, and it also heavily relies on the author's personal experience instead of experimental data or peer-reviewed studies, which limits its credibility. (7) I can also use this article as a source to learn PCB design step-by-step, starting with a schematic and eventually reaching the manufacturing stage. Combining this with other sources will give me a deeper understanding of how routing, power, and heat all fit into the overall design process. (8) This is very relevant to my ISM journey, as the main topic I am studying is PCB design and my final goal is to make one of my own PCBs, and this source helps me understand what and how to develop the skills I need to eventually accomplish my personal goal.

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(1) Biner, George. "Tips and Techniques for DC-DC Buck Converter PCB Layout."

Allaboutcircuit, All About Circuits, 3 July 2019,

<https://www.allaboutcircuits.com/technical-articles/techniques-dc-dc-buck-converter-PCB-layout/>. Accessed 11 Sept. 2026.

(2) This article explains how the performance, efficiency, noise, and thermal behavior of a PCB can be affected by layout choices, specifically DC-DC buck converter PCBs. (3) The goal is to teach proper buck-converter PCB design using electrical principles, examples, and actual recommendations from engineers. (4) The article's scope is on paths for current, trace routing, grounding, placing components, PCB layers, and heat management in buck-converter circuits. (5) This source helped me realize how and in what ways the real, physical layout of a PCB can significantly affect how a circuit performs compared to a digital test bench, making me more interested in how real engineers go through this process. (6) Some limitations are that the article was published in 2019, so some recommendations might not represent recent developments in PCB design; it also doesn't provide much experimental data or comparisons between layouts. (7) I can use the knowledge to look further into how different PCB layouts affect the performance of circuits meant to take on loads of power, helping me design my own PCB with the ideas and concepts I learnt. (8) This is very relevant to my ISM journey because it shows me that designing a PCB is much more than just routing stuff together and that there is a certain level of understanding required to make electricity move efficiently through a board.

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(1) Embedded Staff. "The Basics of Printed Circuit Board Design." *Embedded*, 10 Dec. 2013, <https://www.embedded.com/the-basics-of-printed-circuit-board-design/>. Accessed 11 Sept. 2026.

(2) This article introduces the basics of PCB construction, materials, layers, and manufacturing files and explains what a PCB is made of and the general process of making one. (3) The purpose of the article is to give beginners foundational PCB knowledge through explanations and diagrams, since it's an educational technical source rather than original research, so it focuses on going through established topics rather than new experimental findings. (4) The article focuses on content the other articles talk about as well as new information such as substrates, stackups, and manufacturing files, which are areas that are essential to understanding the physical structure of a board, which is needed to design more complicated PCBs. (5) The article helped me understand what the physical components that make up a PCB board are and why different layers are needed for power, routing, and grounding, giving me a stronger foundation for understanding more advanced PCB topics from other articles. (6) Although the article provides a great foundation, it is mainly introductory and leaves out a lot of modern and advanced PCB design topics, such as newer high-speed signalling and more advanced manufacturing techniques; additionally, it's an older publication date and is therefore less useful for evaluating current PCB technology and practices. (7) I can also use this as a foundation to help me understand routing, power, and thermal management to better understand complete PCB design, giving me a stronger starting point for eventually making my own PCB. (8) This is relevant to my ISM journey because it gives me the basic knowledge I need before studying more advanced boards and microcontrollers, which will help me see what goes into creating a custom board.